

ML Models Best Suited for Different Dataset Types

1. Tabular Data

Situation	Recommended Models
Small/medium structured data	XGBoost, LightGBM, CatBoost
Many categorical columns	CatBoost
Need interpretability	Logistic Regression, Decision Tree
Nonlinear relationships	XGBoost, Random Forest

2. Image Data

Task	Recommended Models
Image classification	ResNet, EfficientNet, Vision Transformer
Object detection	YOLO, Faster R-CNN
Segmentation	U-Net, Mask R-CNN
Real-time detection	YOLOv8/YOLO11

3. Text / NLP Data

Task	Recommended Models
Text classification	BERT
Chatbots	GPT-style transformers
Summarization	T5, BART
Semantic search	Sentence Transformers

4. Time Series Data

Situation	Recommended Models
Classical forecasting	ARIMA, Prophet
Seasonal patterns	Prophet
Long sequential data	LSTM, Transformers
Industrial forecasting	XGBoost with lag features

5. Audio Data

Task	Recommended Models
Speech-to-text	Whisper
Audio classification	CNN + spectrogram
Voice synthesis	Tacotron, VITS

6. Recommendation Systems

Dataset Type	Recommended Models
User-item interactions	Matrix Factorization
Large-scale recommendation	DeepFM, Two-Tower models
Sequential recommendation	Transformers

7. Dataset Size Guidelines

Dataset Size	Preferred Models
<10k rows	Tree models
10k–1M rows	XGBoost / LightGBM
Millions+ rows	Deep Learning
Very little labeled data	Transfer Learning

Key Practical Advice

Tabular Data: Gradient boosting models like XGBoost and LightGBM usually outperform deep learning.

Images, Audio, NLP: Deep learning models dominate performance.

Real-world ML success depends more on:

1. Data quality
2. Feature engineering
3. Label quality
4. Problem framing
5. Model selection